Greg Emmen

Senior Component HIL Simulation and Integration Engineer

(248) 495-0680
greg.emmen@gmail.com
South Lyon, MI
linkedin.com/in/GregEmmen

Professional Experience

General Motors, Senior Component HIL Simulation and Integration Engineer, *Mar 2025 – Present*

Architect project solutions and deploy feature functionality for customers in Hardware-in-the-Loop (HIL) labs
Build tools (python JavaScript, MATLAB, Simulink) to support HIL development workflow
Maintain lab capability (hardware, instrumentation, testing ecosystem)
Develop solutions for validating and testing automotive ethernet
Operate as a scrum lead for a team of engineers

General Motors, Vehicle Systems Integration Engineer for GM Defense, *Feb 2024 – Mar 2025*

Manage the development and integration of electrical systems into defense vehicles
Collaborate with GM Defense team members to maintain and pursue defense contracts
Ensure that integrated vehicle systems comply with regulatory standards and requirements
Identify and resolve issues related to vehicle system integration

General Motors, Software & Controls Engineer for GM Defense, *Sep 2021 – Feb 2024*

Executed mid-cycle hardware and software update to avoid production stoppage due to parts shortage
Prototyped control systems for military vehicles to meet customer requirements with tight deadlines
Utilized real-time CAN communication tools (neoVI, ETAS INCA/MDA) to troubleshoot vehicle issues
Coordinated with suppliers on hardware and software integration for off-the-shelf components
Performed hardware validation activities with dyno cells and mule vehicles

General Motors, Controls Design Engineer, *Dec 2018 – Sep 2021*

Led a cross-functional team of engineers to enable HV propulsion capability and safety features
Defined a leader/follower communication protocol for coordinating actions across controllers
Refactored C++ code to improve run-time execution, maintainability, and software quality
Directed the automation of unit and behavioral software tests with global team
Engaged in GM standard system analysis of failure modes (DFMEA)

General Motors, Plant Modeling & HIL Integration Engineer, *Jun 2015 – Dec 2018*

Technical expert on the implementation of controller software to HIL benches
Built and maintained MATLAB plant models to simulate vehicle hardware capability
Managed multiple HIL benches to meet customer requirements for testing
Enabled Python scripted test automation capabilities for HIL benches

Education

University of Michigan – Dearborn, Master of Science in Engineering (Energy Systems Engineering), *2019 – 2021*

- Regenerative Braking Control of a Brushless DC Motor Drive
- Satellite Attitude Control Model

Colorado State University, Bachelor of Science in Mechanical Engineering, *2011 – 2015*

- CSU EcoCAR3 Controls Team

Skills & Capabilities

Project Coordination	Problem Solving	Agile Development	Programming (Python, C++)
Communication	Adaptability	Data Analysis	MATLAB/Simulink